

Full Stack Development Summary

What is Exception and its types .

Multiple catch blocks .

Finally block .

Difference between throw and throws.

Custom Exception.

Lambda Function.

WHAT IS EXCEPTION AND ITS TYPES

- + Exception is a abnormal event that occurs during the execution of a program.
- + Disrupts the normal flow of instructions.
- + Exceptions can be handled so that the program does not terminate unexpectedly.

TYPES OF EXCEPTION

- + Checked exception.
- + Unchecked exception.
- + Errors.

CHECKED EXCEPTION

- + These are exceptions checked and enforced by the compiler.
- + You must to handle them using try-catch and throws.

EXAMPLES

- + **IOException**
- + **SQLException**

- + **ClassNotFoundException.**
- + **FileNotFoundException**

UNCHECKED EXCEPTION

These are occurs during the runtime and are not checked by the compiler.

EXAMPLE

- + **ArithmeticException.**
- + **NullPointerException.**
- + **ArrayIndexOutOfBoundsException.**
- + **NumberFormatException.**

ERRORS

- + **Errors are serious issues that are not meant to be handled by the applications.**

EXAMPLE

- + **OutOfMemoryError**
- + **StackOverflowError**
- + **VirtualMachineError**

MULTIPLE CATCH BLOCKS

- ✚ A multiple catch block means using more than one catch block with a single try block to handle different types of exceptions separately.
- ✚ A single try block they can have multiple try blocks.
- ✚ The most specific exceptions should come before the more general exception.
- ✚ The exception class should usually be the last catch block.

FINALLY BLOCK

- ✚ A finally block is used to execute code whether an exception occurs or not
- ✚ Its mainly used for cleanup operations such as closing files,database connections or releasing resources.
- ✚ The finally block always execute, whether an exception occurs or not.
- ✚ It is optional and used for cleanup the code.

DIFFERENCE BETWEEN THROW AND THROWS

Throw	Throws
Used to explicit throw an exception	Used to declare exception that a method may throw
Used inside a method	Used in the method signature
Throws one exception at a time	Can declare multiple exceptions

CUSTOM EXCEPTION

- ✚ A custom exception is a user defined exception created by the programmer to handle the applications -m specific errors.
- ✚ It is created by extending the exception class or Runtime Exception class.
- ✚ To handle applications specific errors.
- ✚ Makes the code more readable.

LAMBDA FUNCTION

- ✚ A lambda expression is a short way to write an anonymous function
- ✚ It was introduced in java 8 to make code shorter and easier to read.